

# PATTERN RECOGNITION AND MACHINE LEARNING

## Lecture 09 – Support Vector Machines for Classification and Regression

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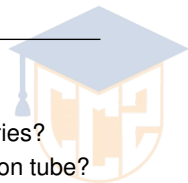
Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences



Fall 2026

| Syllabus block            | Time  | Driving question                                    |
|---------------------------|-------|-----------------------------------------------------|
| Linear SVM                | 0.5 h | Which separating line is safest?                    |
| Soft margin               | 0.5 h | How much violation should be tolerated?             |
| Kernel methods            | 1.0 h | How can similarity create nonlinear boundaries?     |
| Support vector regression | 1.0 h | Which residuals should constrain a regression tube? |



story from inspection

Factory gate

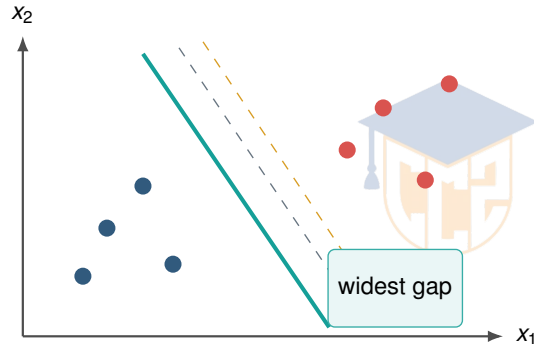
Two acceptable product families can be separated by several straight lines.

Tomorrow

A measurement may shift slightly after sensor recalibration.

Choice

A boundary with no clearance can fail after a tiny shift.



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- 1 Choose a geometric separation.
- 2 Permit measured violations when data are noisy.
- 3 Rewrite the solution with sample similarities.
- 4 Replace inner products by kernels.
- 5 Use an epsilon tube for regression.

## A hyperplane separates two classes

$$f(x) = w^T x + b, \quad \hat{y} = \text{sign}(f(x))$$

## Read it

A two-feature defect dataset becomes a line on the screen.

## Boundary

The score is not yet a calibrated probability.

# The weight vector is normal to the boundary

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## Core idea

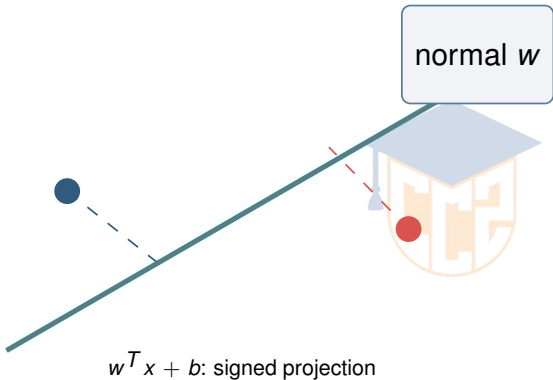
$w$  points perpendicular to  $w^T x + b = 0$ .

## Concrete case

Project a sample onto the normal to measure its signed position.

## Watch for

Reading coefficient size without feature scaling distorts the geometry.





## Labels make one signed constraint

$$y_i(w^T x_i + b) > 0, \quad y_i \in \{-1, +1\}$$

Read it

Both classes satisfy the same inequality after multiplication by the label.

## Boundary

A zero score lies exactly on the decision boundary.



# Parameter rescaling fixes the functional margin

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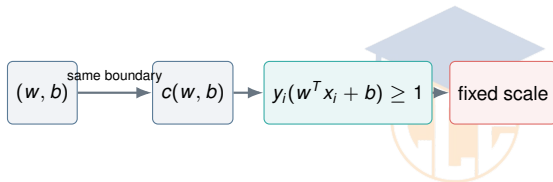
$$y_i(w^T x_i + b) \geq 1$$

## Read it

Scale a separating solution until the closest case reaches one.

## Boundary

The number one is a normalization convention, not a physical distance.





# The full margin width is two over the norm

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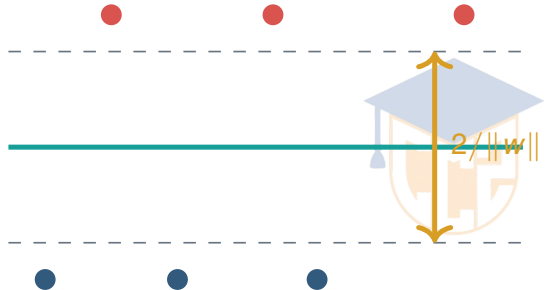
$$\rho = \frac{2}{\|w\|_2}$$

Read it

A smaller weight norm produces a wider corridor after normalization.

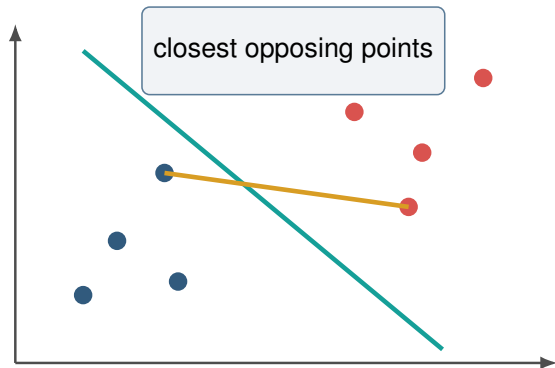
Boundary

The formula depends on the fixed functional-margin convention.



One contradictory or mislabeled case makes the problem infeasible.

## Only the closest cases set the solution



## Start from evidence

Move a point that touches a supporting plane and the boundary may rotate.

## Model statement

Moving a distant point slightly does not change the optimum.

## Failure mode

A visually extreme point is not necessarily a support vector.

# Support vectors encode local boundary evidence

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## What changes

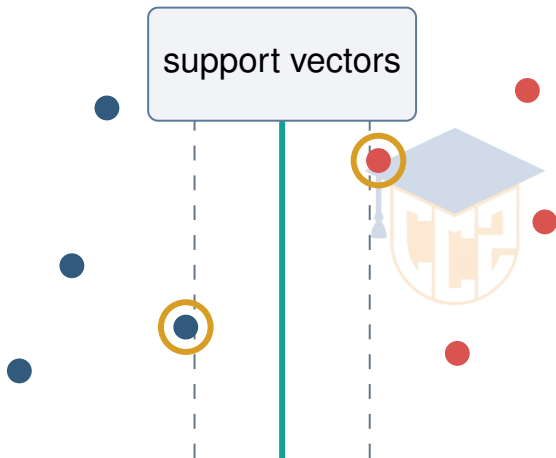
The classifier is determined by a subset near the class frontier.

## What can break

Small support count does not automatically imply good generalization.

## Try this

A large dataset can yield a small support set.



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Blue cases lie left of three and red cases lie right of three.

Many scaled parameter pairs represent the same line.



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## Several correct lines are not equally robust

## Start from evidence

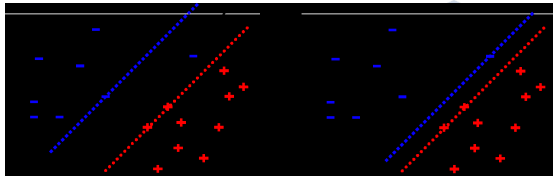
Compare the minimum signed distance, not the average distance.

## Model statement

Zero training error does not distinguish separators on separable data.

## Failure mode

A line that is far from most points may still pass dangerously close to one case.





| Lens      | Evidence                                                                      |
|-----------|-------------------------------------------------------------------------------|
| Principle | There exists at least one hyperplane satisfying every unit-margin constraint. |
| Case      | Clean synthetic points may meet this condition.                               |
| Risk      | Real labels, overlapping classes, and measurement errors often violate it.    |

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has  $0 < \xi_i < 1$ .

A correctly classified point inside the corridor has  $0 < \xi_i < 1$ .

Slack is not simply the count of mistakes.

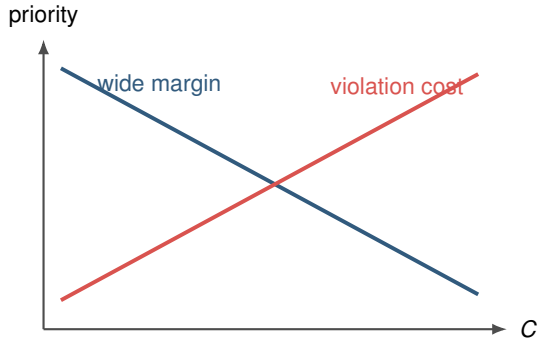


100

100

Large  $C$  is not always better.

## The parameter C sets the violation price



## Core idea

Small  $C$  favors a wider margin; large  $C$  emphasizes training violations.

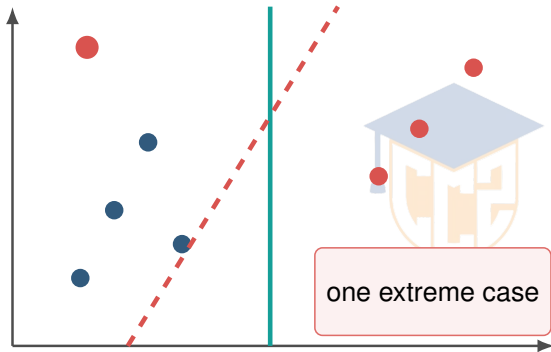
## Concrete case

Trace a validation curve over logarithmic values of  $C$ .

## Watch for

The best value depends on scaling and sample size.





Correct predictions inside the margin still incur loss.



Read it

A safely classified point with  $z > 1$  adds no loss.

## Boundary

A zero hinge loss is stronger than merely being correct.

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## Class weights alter the cost of violations

$$\frac{1}{2}\|w\|^2 + \sum_i C_{y_i} \xi_i$$

Read it

Give rare defect cases a larger violation price.

## Boundary

Weights should reflect costs and validation evidence, not only class counts.

## Duplicate samples also change effective weights

## Core idea

Repeating one observation multiplies its contribution to hinge loss.

## Concrete case

A duplicated customer record can pull the boundary.

## Watch for

Deduplication is part of model validity, not clerical cleanup.

## Regulariz

Boundary

Do not compare raw numbers across conventions.

## Validation chooses C, not training accuracy

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## Start from evidence

Select on the deployment-relevant metric.

## Model statement

Use stratified or grouped cross-validation appropriate to the data source.

## Failure mode

The final test set must remain outside the search.







## Soft-margin boundaries differ by design

## Core idea

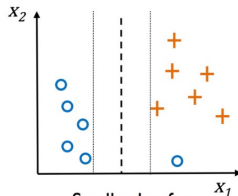
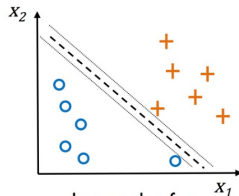
A wider corridor may deliberately accept a few training errors.

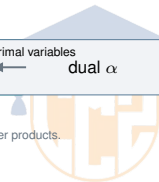
## Concrete case

Compare a hard-looking boundary with a tolerant one on the same points.

## Watch for

Visual simplicity is not proof of better expected performance.











# The dual objective contains only inner products

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$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

## Read it

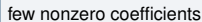
Subject to  $\alpha_i \geq 0$  and  $\sum_i \alpha_i y_i = 0$ .

## Boundary

The Gram matrix can be large when the sample count is large.



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Most cases far from the boundary receive coefficient zero.

Only active constraints have positive multipliers.

Soft margins may produce many bounded coefficients at  $C$ .

A point on the margin may still have a numerically tiny coefficient.

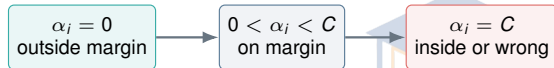
$$0 \leq \alpha_j \leq C$$

## Read it

Coefficients at  $C$  correspond to strongly violating or boundary-limited cases.

## Boundary

Numerical tolerances blur exact equality.



Complementary slackness converts algebra into sample roles.

## Recover the intercept from free support vectors

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$$b = y_i - \sum_j \alpha_j y_j K(x_j, x_i), \quad 0 < \alpha_i < C$$

## Read it

Average the estimate over several free support vectors.

## Boundary

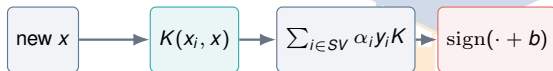
Using a bounded point at  $C$  can bias the estimate.

## Prediction sums similarities to support vectors

$$f(x) = \sum_{i \in SV} \alpha_i y_i K(x_i, x) + b$$

## Read it

A new case compares only with retained support vectors.



## Boundary

Prediction time grows with the support-set size.

Compare one new point with one positive and one negative support.

Similarity magnitudes depend on the chosen kernel and scaling.



## T1

$$y_1\alpha_1 + y_2\alpha_2 = \zeta, \quad 0 \leq \alpha_1, \alpha_2 \leq C$$

## Read it

Intersect an equality line with the coefficient box.

## Boundary

The endpoints differ for equal and opposite labels.



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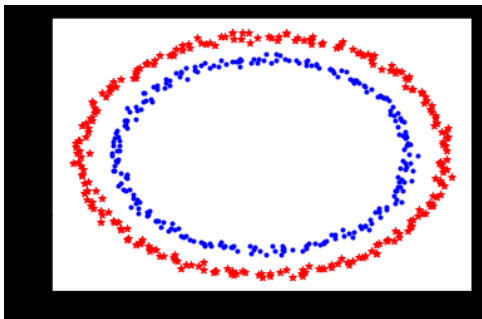
Recheck error-cache values after an update.

## A stale cache produces incorrect priorities.





## A nonlinear pattern can become linearly separable



## Core idea

Map inputs to a feature space where a hyperplane is useful.

## Concrete case

Concentric rings are not separable by a line in the original plane.

## Watch for

The mapped space may be very high or infinite dimensional.

# Feature maps make the hidden geometry explicit

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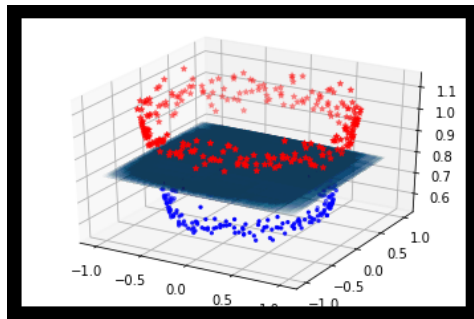
$$\phi : x \mapsto \phi(x)$$

Read it

Use  $\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$  for rings.

Boundary

Manually enumerating rich features can be expensive.



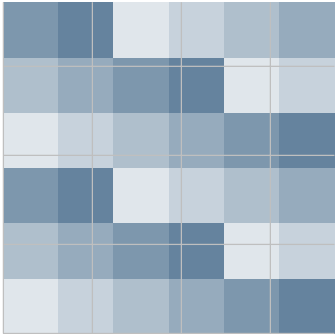
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Compute similarity without materializing every mapped coordinate.

Not every arbitrary similarity function is a valid kernel.

# A Gram matrix stores all training similarities

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symmetric

positive semidefinite

pairwise similarity

$$K_{ij} = K(x_i, x_j)$$

Read it

Rows and columns correspond to the same training cases.

Boundary

Memory grows quadratically with sample count.

## Valid kernels are symmetric positive semidefinite

$$K = K^T, \quad c^T K c \geq 0 \quad \forall c$$

Read it

A positive semidefinite Gram matrix defines a consistent inner-product geometry.

## Boundary

A hand-designed score may violate this property.



## Kernel sums preserve validity

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$$K = aK_1 + bK_2, \quad a, b \geq 0$$

## Read it

Combine text and metadata similarities with nonnegative weights.

## Boundary

Negative weights can destroy positive semidefiniteness.

It gives the ordinary linear SVM in dual form.

A linear kernel does not eliminate the need for scaling.



## An explicit degree-two expansion

$$(X^T Z)^2 = \langle \phi(X), \phi(Z) \rangle$$

## Read it

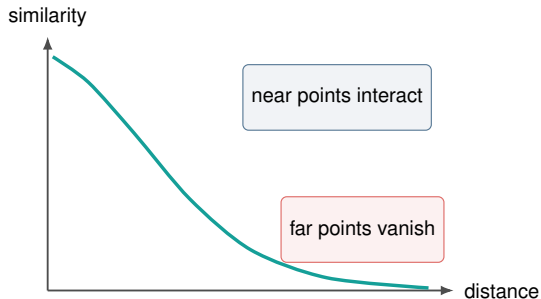
For two inputs, the map contains squared terms and a scaled cross term.

## Boundary

The square-root scaling on cross terms matters for exact equality.

# The RBF kernel measures local similarity

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$$K(x, z) = \exp(-\gamma \|x - z\|_2^2)$$

## Read it

Nearby points have similarity near one; far points approach zero.

## Boundary

Distance is meaningless if feature scales are incompatible.

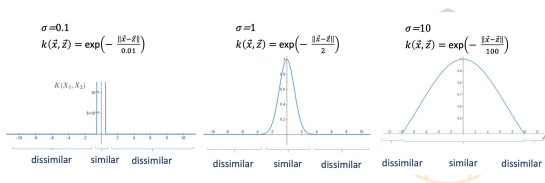
$$\gamma = \frac{1}{2\sigma^2}$$

Read it

Larger  $\gamma$  means a narrower influence region.

## Boundary

Parameter names differ across libraries and texts.



# Small gamma produces smooth boundaries

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## Core idea

Many distant points contribute appreciable similarity.

## Concrete case

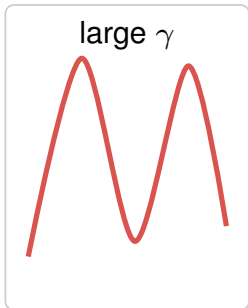
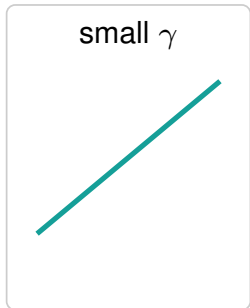
Use it when class structure changes gradually in scaled space.

## Watch for

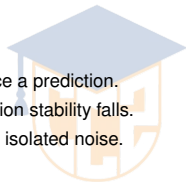
Too small a value can underfit local structure.

# Large gamma produces local islands

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- Only very near support vectors influence a prediction.
- Training accuracy can rise while validation stability falls.
- A fragmented boundary may memorize isolated noise.



Kernel width controls the locality of the boundary.



# C and gamma interact

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## Start from evidence

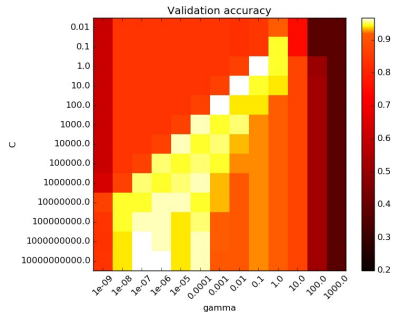
A ridge of good validation scores may run diagonally across the grid.

## Model statement

High  $C$  and high  $\gamma$  can jointly create a highly flexible classifier.

## Failure mode

Tuning one parameter while fixing the other can miss the useful region.



## Read a C-gamma heatmap

## What changes

Each cell is a cross-validated score for one parameter pair.

## What can break

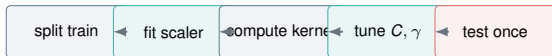
The brightest cell may be a noisy winner.

Try this

Prefer a stable plateau when scores are practically tied.

# Nested validation protects the final estimate

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Every transformation is learned inside the training fold.

## Core idea

Tune kernel settings inside the inner loop and assess the whole procedure outside.

## Concrete case

Every outer fold repeats scaling and parameter search.

## Watch for

Reusing outer-test feedback leaks model-selection information.

# A movie similarity story

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## Core idea

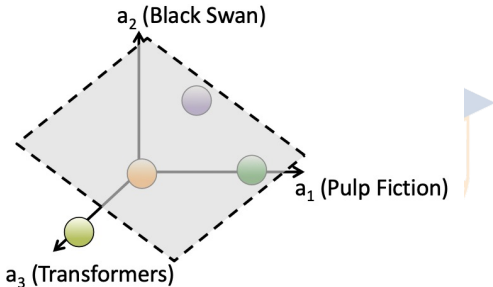
Represent each film with budget, rating, and genre-related features.

## Concrete case

A Palme d'Or winner may be surrounded by films that are locally similar.

## Watch for

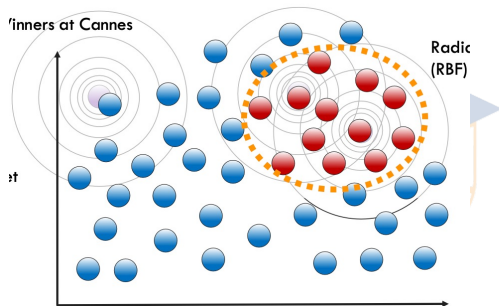
Raw feature distances mix units and semantics.



# RBF features place local bumps around anchors

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- Each anchor contributes a radial response centered on itself.
- Several anchor responses can separate a curved group of films.
- Too many narrow bumps can memorize training cases.



# Kernel centering matters for some methods

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$$K_c = HKH, \quad H = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$$

## Read it

Center feature-space coordinates without constructing them explicitly.

## Boundary

Fit centering statistics only on training data when used in a pipeline.

# Kernel normalization controls self-similarity

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$$\tilde{K}(x, z) = \frac{K(x, z)}{\sqrt{K(x, x)K(z, z)}}$$

## Read it

Useful when vector magnitudes should not dominate direction-like similarity.

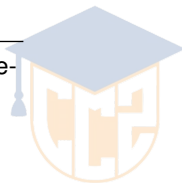
## Boundary

Normalization changes the model and must be validated.

# Categorical inputs need an explicit design

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| Lens      | Evidence                                                      |
|-----------|---------------------------------------------------------------|
| Principle | One-hot encoding plus a linear or RBF kernel is one baseline. |
| Case      | A custom equality kernel can compare category matches.        |
| Risk      | Integer category codes create fake distances.                 |





# Text often favors sparse linear SVMs

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## Core idea

High-dimensional sparse counts can already be linearly separable enough.

## Concrete case

TF-IDF with a linear SVM is a strong document baseline.

## Watch for

Dense RBF Gram matrices may be unnecessary and expensive.

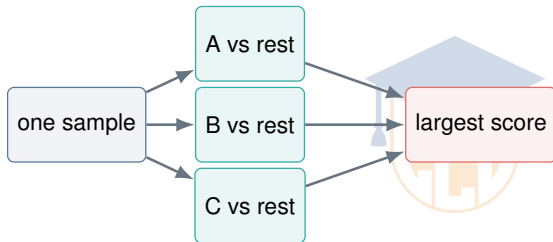
## Image descriptors may benefit from nonlinear similarity

- Kernel choice should reflect the invariances encoded by the representation.
- Compare linear and RBF kernels after a fixed feature extractor.
- A better kernel cannot repair a representation that discards the signal.

Choose the class with the largest comparable decision score.

Class  $k$  is positive and all remaining classes are negative.

Independent binary scores are not calibrated probabilities by default.



# One-vs-one builds pairwise classifiers

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$$\frac{K(K-1)}{2} \text{ binary models}$$

## Read it

Each pair votes or contributes a coupled score.

## Boundary

The number of models grows quadratically with class count.







# Kernel matrix memory grows quadratically

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$$N = 50,000 \Rightarrow N^2 = 2.5 \times 10^9 \text{ entries}$$

## Read it

At eight bytes per entry, the raw matrix is about twenty gigabytes.

## Boundary

Solver overhead adds more memory beyond the matrix.



## A short kernel experiment

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

X, y = make_classification(n_samples=320, n_features=4, random_state=7)
X_train, X_valid, y_train, y_valid = train_test_split(X, y, random_state=7)
model = make_pipeline(StandardScaler(), SVC(kernel='rbf', C=10, gamma=0.1))
model.fit(X_train, y_train)
print(model.score(X_valid, y_valid))
```

## Expected observation

Compare linear and RBF validation scores under the same split.

## Check

Do not scale the validation data separately.

## Inspect support vectors after fitting

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```
svc = model.named_steps['svc']
print(svc.n_support_)
print(svc.support_.shape)
```

## Expected observation

Report support counts by class and total support fraction.

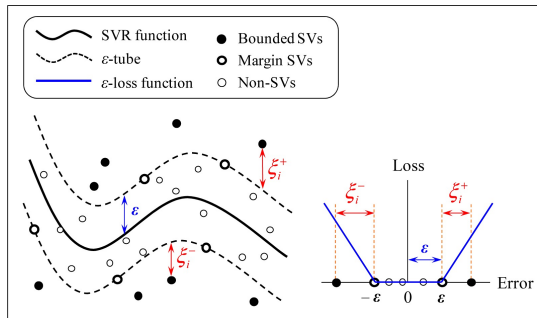
## Check

Indices refer to the transformed training order.

# Regression replaces a margin corridor with a tube

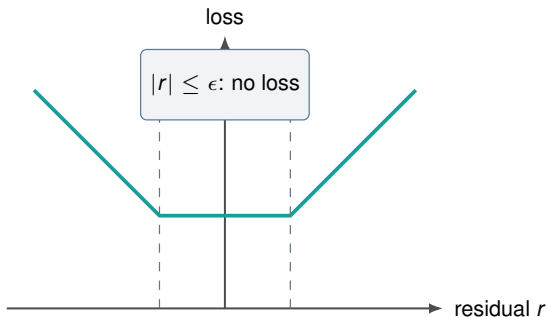
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- Fit a function so that many targets lie within  $\pm\epsilon$ .
- Small deviations inside the tube are ignored.
- The tube width is a modeling tolerance, not a confidence interval.



# Epsilon-insensitive loss has a flat center

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$$L_{\epsilon}(r) = \max(0, |r| - \epsilon)$$

## Read it

A residual of 0.3 pays no loss when  $\epsilon = 0.5$ .

## Boundary

Residuals outside the tube pay only the excess beyond  $\epsilon$ .

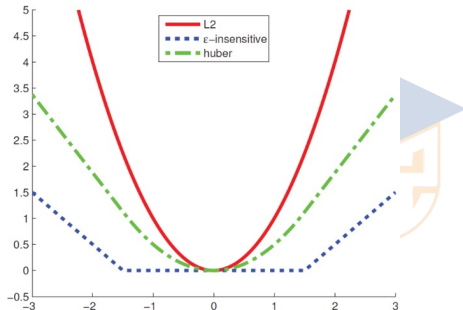
Squared loss grows quadratically; Huber changes regime; epsilon loss ignores a central band.

## What can break

Losses with similar curves near zero can behave very differently on large residuals.

## Try this

Choose the loss from the error cost, not from habit.



## Linear SVR primal objective

$$\min_{w, b, \xi, \xi^*} \frac{1}{2} \|w\|^2 + C \sum_i (\xi_i + \xi_i^*)$$

Read it

Use separate slacks above and below the tube.

## Boundary

The two sides need not have different costs unless the task requires asymmetry.



# Support vectors sit on or outside the tube

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- Points strictly inside the tube have zero effective dual coefficient.
- Boundary and outside points determine the fitted function.
- A target can be a support vector without being an extreme outlier.





## Dual SVR uses coefficient differences

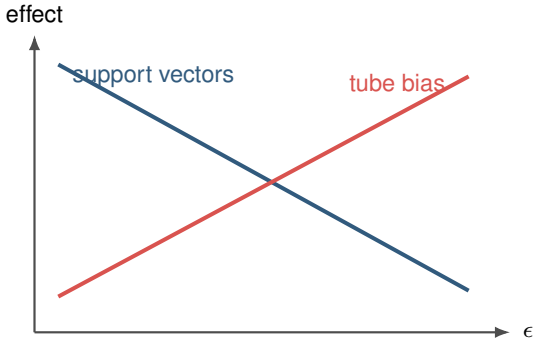
$$f(x) = \sum_i (\alpha_i - \alpha_i^*) K(x_i, x) + b$$

Read it

Upper and lower violations contribute with opposite signs.

## Boundary

Both coefficient families are bounded by  $C$ .



## C controls outside-tube error

| Lens      | Evidence                                                             |
|-----------|----------------------------------------------------------------------|
| Principle | Large $C$ bends the function more strongly toward violating targets. |
| Case      | Compare validation MAE and support fraction across $C$ .             |
| Risk      | High $C$ with high gamma can overfit individual targets.             |

## Target scaling changes epsilon meaning

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- Scaling  $y$  requires scaling or retuning  $\epsilon$  and interpreting metrics back in original units.
- An epsilon of 0.1 means different things for standardized and raw demand.
- Do not compare raw epsilon across target transformations.

## SVR is not safe for extrapolation

## Start from evidence

Future temperatures beyond the training range have weak similarity to every support vector.

## Model statement

RBF predictions far from all training cases tend toward the intercept-like baseline.

## Failure mode

A smooth validation curve inside range does not prove extrapolation reliability.





Keep scaler fitting inside every training fold.

The default score is  $R^2$ , not MAE.





## Sample size changes the visible margin

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## What changes

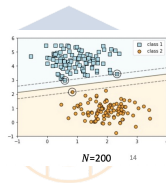
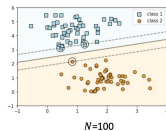
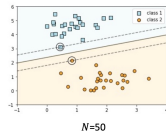
With more observations, the estimated class frontier becomes better constrained.

## What can break

More duplicated or biased data do not improve representativeness.

### Try this

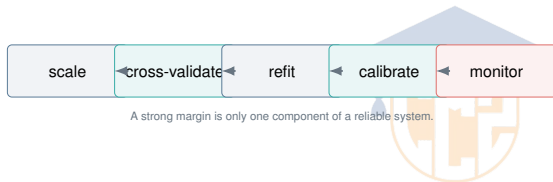
Compare the same data-generating process at  $N = 50, 100$ , and  $200$ .



## A classifier comparison must share evidence

| Lens      | Evidence                                                              |
|-----------|-----------------------------------------------------------------------|
| Principle | Use identical splits, preprocessing, metrics, and search budgets.     |
| Case      | Compare linear SVM, RBF SVM, and logistic regression on one pipeline. |
| Risk      | Giving one model a larger tuning budget biases the contest.           |





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